Can Agentic AI Settle a 20-Year-Old Open Problem?

The decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ — a machine-checked campaign

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The one-slide version

- In 2002/2003 I defined a family of spatial description logics and left their central decidability question open. It still is.
- In 2026 I directed a four-month AI campaign on it: two frontier models, adversarial review, machine verification.
- The question is not settled. But it is now *reduced* — honestly and partly kernel-checked — to one well-posed lemma, with a decidable fragment, a certified normal form, a working reasoner, and an exact arithmetical address (Π^0_1) along the way.

How to read everything in this talk
The mathematics was produced by AI assistants and is *unverified by human experts*. Standing label: **strongly supported, *not* certified** — except the parts checked by the Lean 4 kernel, which are machine-verified (but still human-unreviewed).

Five ways two regions can relate (RCC5)

DR	PO	EQ	PP	PPI
disjoint	partial overlap	equal	proper part	proper superpart

- Every pair of regions stands in exactly one of these five relations.
- *Jointly exhaustive, pairwise disjoint* — the classic qualitative spatial calculus (a coarser cousin of RCC8).
- Think GIS, layout, biology, any "what contains / touches what" domain — no coordinates required.

Reasoning without geometry: the composition table

If I know $x\ R\ y$ and $y\ S\ z$, what can $x\ ?\ z$ be?

known	known	forced / allowed
$x\,\mathrm{PP}\,y$	$y\,\mathrm{PP}\,z$	$x\,\mathrm{PP}\,z$ (*forced*: parts of parts are parts)
$x\,\mathrm{PP}\,y$	$y\,\mathrm{DR}\,z$	$x\,\mathrm{DR}\,z$ (*forced*: inside something disjoint)
$x\,\mathrm{DR}\,y$	$y\,\mathrm{DR}\,z$	anything at all (*free*: 5 options)

- All 25 cells form the composition table — pure algebra, derived once from set semantics, then used forever.
- Abstract semantics: a “model” is any relation assignment closed under this table — *space itself has left the building*.
- Every probe in this project re-derives the table from scratch, so the algebra can never silently drift.

Description logic in one slide

Concepts = definitions built from properties and quantified relations:

$$\mathsf{GermanCity} \equiv \mathsf{City} \sqcap \exists \mathsf{PP}.\,\mathsf{Germany} \qquad \text{“a city \textit{properly inside} Germany”}$$

$$\mathsf{CityAtRiver} \equiv \mathsf{City} \sqcap \exists \mathsf{PO}.\,\mathsf{River} \qquad \text{“a city \textit{overlapping} some river”}$$

A reasoner then derives the taxonomy *automatically*:

$$\mathsf{Hamburg} \sqsubseteq \mathsf{GermanCity} \quad \text{and} \quad \mathsf{Hamburg} \sqsubseteq \mathsf{CityAtRiver}.$$

- The decision problem underneath everything: concept satisfiability — “can this definition be realized at all?”
- Subsumption, classification, consistency all reduce to it.

$\mathcal{ALCI}_{\mathrm{RCC5}}$: the spatial relations *become* the roles

The logic (Wessel 2002/2003):

- $\mathcal{ALCI}$ = the basic DL with inverse roles;
- roles = the five RCC5 relations themselves;
- the composition table is enforced globally;
- EQ = identity (*strong* semantics).

So one can say things like

$$\forall \mathsf{PP}.\, \neg \mathsf{Lake} \qquad \exists \mathsf{DR}.\, \mathsf{Reserve}$$

"no part is a lake", "disjoint from some reserve".

Why this is different
Concrete-domain DLs constrain spatial *attributes* of objects. Here the relations are *first-class roles*: one can quantify over them. None of the known concrete-domain decidability results apply.

Why the standard toolbox dies on contact

Every pair of elements carries a relation $\Rightarrow$ a model is a complete labelled graph, never a tree.

no tree models

Blocking/unravelling — the workhorse of DL decidability — has no tree to work on.

no finite models

$\exists \mathrm{PPI}.\top \sqcap \forall \mathrm{PPI}.\exists \mathrm{PPI}.\top$ forces an infinite ascending part-of tower.

nondeterministic table

Most cells allow several outcomes — composition *constrains* but rarely *determines*.

Creating one new element forces relations to *every* existing element — and a dormant $\forall$ anywhere may wake up because of it.

The landscape: open exactly at the top

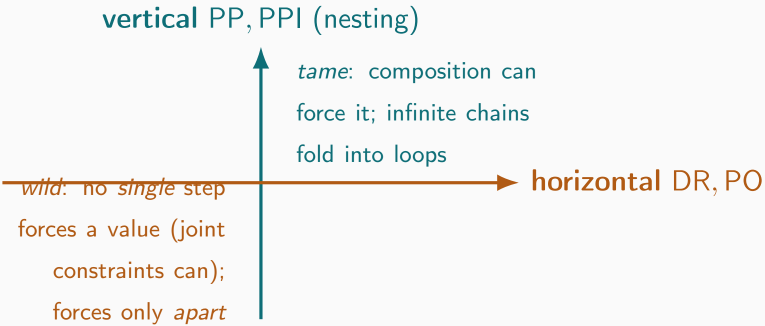
Logic	Relations	Concept satisfiability
$\mathcal{ALCI}_{\mathrm{RCC1}}$	$\{\mathsf{SR}\}$	decidable ($\equiv$ S5)
$\mathcal{ALCI}_{\mathrm{RCC2}}$	$\{\mathsf{DR}, \mathsf{O}\}$	decidable
$\mathcal{ALCI}_{\mathrm{RCC3}}$	$\{\mathsf{DR}, \mathsf{ONE}, \mathsf{EQ}\}$	decidable (via two-variable FO!)
$\mathcal{ALCI}_{\mathrm{RCC5}}$, $\forall$PO-free	RCC5 w/o $\forall$PO	**decidable** (this project)
$\mathcal{ALCI}_{\mathrm{RCC5}}$	all five	**open**; PSpace-hard
$\mathcal{ALCI}_{\mathrm{RCC8}}$	eight RCC8 rel.	**open**; ExpTime-hard

Lutz & Wolter 2006, after proving the *topological* cousins undecidable:

> *"Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [$=\mathcal{ALCI}_{\mathrm{RCC5}}$] … to which the reduction exhibited in Section 8 does not apply."*

Recognized open by the leading experts — for twenty years.

One logic, two completely different axes



- Depth (nesting) is never the problem — ordinary modal-logic folding handles it.
- Everything hard lives in horizontal crowds: many pairwise-distinct neighbours that no finite summary seems to capture.

The asymmetry, made precise (machine-verified)

No *single* step forces a horizontal value. Of the 16 compositions of proper relations, exactly four return a single value:

$$\mathrm{comp}(\mathrm{DR},\mathrm{PPI}) = \mathrm{comp}(\mathrm{PP},\mathrm{DR}) = \{\mathrm{DR}\}, \quad \mathrm{comp}(\mathrm{PP},\mathrm{PP}) = \{\mathrm{PP}\}, \quad \mathrm{comp}(\mathrm{PPI},\mathrm{PPI}) = \{\mathrm{PPI}\}$$

— every one runs through PP/PPI; no single cell yields $\{\mathrm{PO}\}$. But the *intersection* of several can force PO ($\mathrm{comp}(\mathrm{DR},\mathrm{PO}) \cap \mathrm{comp}(\mathrm{PPI},\mathrm{PO}) = \{\mathrm{PO}\}$) — so "shadow" is joint closure, not single-path.

Disjointness is half-deterministic.

downward: $x\,\mathrm{DR}\,y \Rightarrow$ every part of x is DR from every part of y

upward: a superpart of x may be DR, PO, or PPI to y — free

So one high DR edge dispatches an *arbitrarily wide* crowd of disjointnesses below it, for

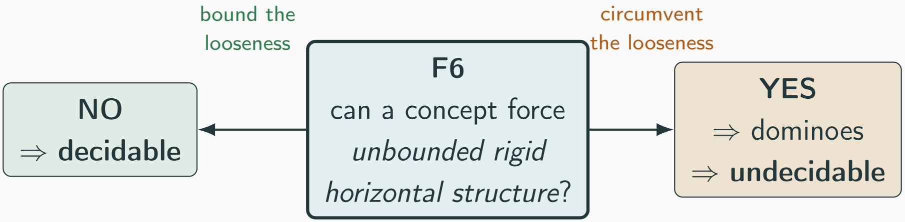
Shadows, live width, and the one property that matters

- An edge is a shadow if composition already forces it from edges you have — it costs a decision procedure *nothing*.
- The live width of an interface = the number of *non-shadow* edges there: the genuine branching that must be recorded.

Property F6 (the keystone of everything)
Is the live width of models bounded by a computable function of the input concept?

If yes $\Rightarrow$ finite certificate space $\Rightarrow$ satisfiability is decidable.

The knife's edge: one fact, two futures



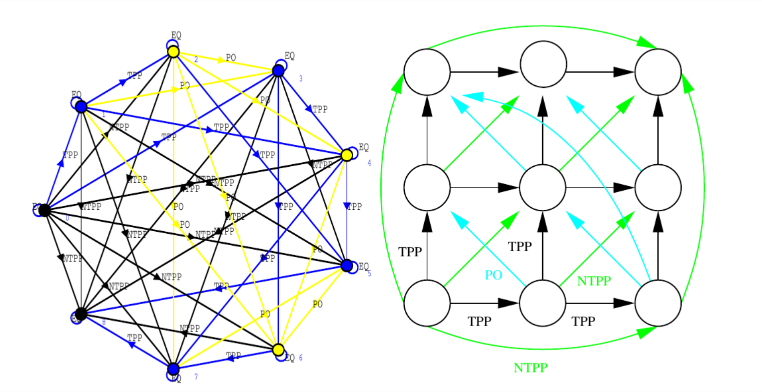
The *same* looseness of the composition table blocks both proofs. That is *why* neither direction has been settled in twenty years — it is one question seen from two sides.

The standard weapon needs a rigid grid

- Classic undecidability proofs encode domino tilings of $\mathbb{N} \times \mathbb{N}$: build a grid, force tile constraints, reduce the halting problem.
- The crux is rigidity: going *east then north* must land on *exactly the same cell* as *north then east* ($R_X \circ R_Y = R_Y \circ R_X$, the "coincidence condition").
- Every known route into $\mathcal{ALCI}_{\mathrm{RCC}}$ needs a feature it does not have: number restrictions, arbitrary role axioms, topological rigidity, nominals. . .

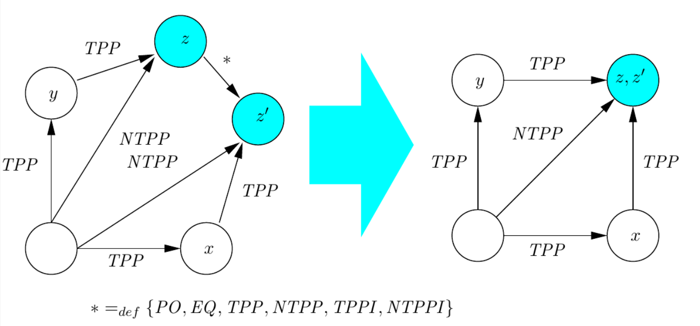
Question (2003 and today): can the composition table itself supply the rigidity?

2003: the grid models *exist*. . .



An RCC8 network isomorphic to an $n \times n$ grid (report7, Fig. 10, 2002/2003) — the construction *found by a Lisp enumerator* and drawn with CLIM (MCL), not by hand. The models are *out there*.

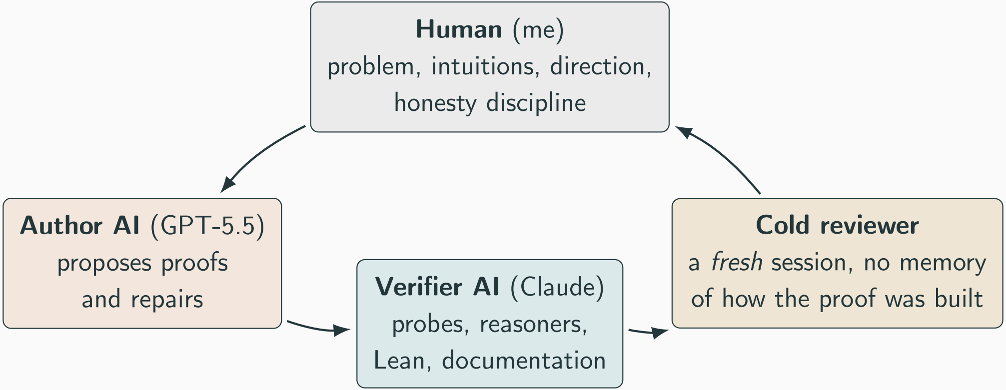
. . . but no concept can *force* them: the coincidence obstruction



report7, Fig. 11: the two successors z (east-north) and z' (north-east) *should* be the same corner — but the table allows six relations between them. Leave it loose: the branches never join. Force a merge with isolation markers: *every* corner collapses to one point (verified again in 2026, at depth 2, for RCC5 *and* RCC8).

> *"Even though these infinite models do exist, we have found no way to enforce the depicted model(s) by means of a concept term."* (report7, 2003)

The cast, and the loop



Propose $\rightarrow$ verify $\rightarrow$ attack $\rightarrow$ repair. Round after round, March–July 2026.

The campaign at a glance

- ∼4 months, ∼30 repair rounds
- 17 adversarial reviews — a defect or overclaim found in *all but two*
- 88 self-contained verification probes (each re-derives the algebra from set semantics)
- 4 Lean files, ∼5,900 lines, zero `sorry`
- 2 working reasoners, cross-validated on 911 concepts

The single most telling number
Across *all* rounds and reviews, no accepted-but-unsatisfiable concept was ever found. Every defect was a gap in an *argument* — never a counterexample to the intended theorem.

That is why the label is *strongly supported* — and why it is still *not certified*.

Route 1: tableaux with blocking — broken by totality

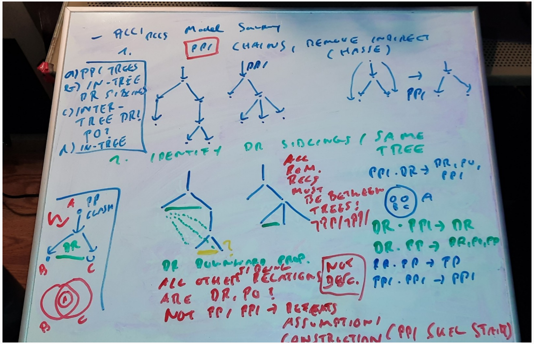
The default DL method: build a model top-down, *block* when a node repeats an ancestor's type.

- Here, composition forces edges into already-built nodes: from $x\,\mathrm{PP}\,y$, $y\,\mathrm{DR}\,z$ the table forces $x\,\mathrm{DR}\,z$.
- Diagnostic: $C_{\text{force}} = \exists\mathrm{PP}.(\exists\mathrm{DR}.A) \sqcap \forall\mathrm{DR}.\neg A$ is **unsatisfiable** — any procedure that ignores forced edges wrongly accepts it.
- Dormant $\forall$'s reawaken blocked nodes; no fixed-arity local summary (types, triangles, quadruples, profiles, ...) is ever enough.

Lesson: in a total-relation logic, "local" does not exist.

What may be reused depends on the whole graph. (That is F6 in disguise.)

Route 2: the split-forest idea (a whiteboard, April 2026)



- Present the complete graph as a *forest of part-of trees*: split shared parts into EQ-mates, re-orient vertical edges into the trees.
- All remaining cross-edges are horizontal; complete them by the RCC5 *patchwork property* (a citable classical theorem).
- The one component every cold review found sound — the semantic foundation all later routes consume.

The human contribution: this decomposition. The models made it precise.

Route 3: tree automata — "acceptance without validity"

Package the search as a parity tree automaton (emptiness is decidable)...

- ... but the input tree omits the relations between fresh witnesses and the sources of $\forall$-constraints.
- An *alternating* automaton explores in independent copies — and copies guess the missing relations independently.
- A cold review built the killer: two copies privately guess $\{\mathrm{PP}\}$ and $\{\mathrm{DR}\}$ for the same pair; $\mathrm{comp}(\mathrm{PP}, \mathrm{PP}) \cap \mathrm{comp}(\mathrm{PP}, \mathrm{DR}) = \{\mathrm{PP}\} \cap \{\mathrm{DR}\} = \emptyset$ — every copy accepts, no model exists.

So the automaton was abandoned, in favour of explicit finite certificates checked by composition tests — with a *truth lemma* instead of an acceptance condition. (This is what got formalized.)

Machine-checking: what the Lean kernel buys

- Lean 4: a proof assistant whose kernel type-checks *every* inference; a proof with a gap *does not compile*.
- Our discipline: Lean 4 core only — no libraries; every object built from first principles; tiny trusted base.
- "`zero sorry`" = no admitted gaps anywhere.

Why we moved to Lean
Most reviews broke a *prose* proof — typically at a step that was *assumed*, not proved. The kernel cannot be talked past. After the move, reviews stopped finding soundness defects and started finding only *interface* gaps — which were then closed.

What exactly is certified (zero `sorry`)

Result	Meaning	File
Soundness pipeline	valid certificate $\Rightarrow$ real RCC5 model	`Round19Transport`
Abstraction faithful	satisfiable $\Leftrightarrow$ Hintikka-realizable	`Round19Transport`
RCC5 normal form (both dir.)	network $\Leftrightarrow$ order + disjointness	`RCC5NormalForm`
Forcing reduction	F6 fails $\Leftrightarrow$ a concept forces crowds	`ForcingReduction`
Dovetail schema	qualitative F6 $\Rightarrow$ decidable	`SemiDecidability`

What is deliberately not in the kernel: the open completeness premise itself (F6). It is *stated*, never axiomatized — the honest boundary between what is proved and what is conjectured.

The main theorem (conditional, soundness certified)

Conditional decidability
If F6 holds — live width bounded by a computable function of the concept — **then** $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability is decidable.

- The soundness half is kernel-checked end to end: certificate syntax $\to$ closure conditions $\to$ proper RCC5 frame $\to$ truth lemma $\to$ *actual model of the input concept*.
- The certificate checker is a real executable Boolean function — no oracle anywhere (a previous version had a hidden one; a cold review caught it; it was rebuilt).
- Everything that remains open is the antecedent — F6.

Unconditional win #1: a decidable fragment

Theorem ($\forall$PO-free fragment)
Concepts with no $\forall$PO subformula are decidable, with live width $\leq (1+m)^2\, 2^{2^m}$, $m = |\mathrm{cl}(C_0)|$.

- Still allowed: $\exists$PO ("overlaps some..."), $\forall$DR, $\exists$DR, and all part-of modalities — a genuinely expressive spatial fragment.
- Why it works: PO is the *only* relation whose recurrence along a chain needs an "all-phase" safety condition; drop $\forall$PO and the condition is vacuous.
- Found twice, independently, by two different routes — a good sign the fragment is natural, not an artifact.

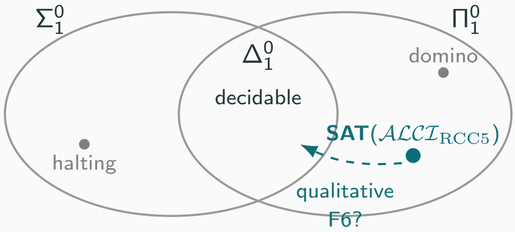
Unconditional win #2: the RCC5 normal form

Theorem (both directions kernel-checked, arbitrary domains)
A strong-EQ RCC5 network is composition-closed *iff* it is an ordered-disjoint structure:

- PP is a strict partial order (a genuine part-of hierarchy);
- DR is a symmetric, downward-closed disjointness;
- comparable pairs never disjoint; PO is the residual.

Every RCC5 model, anywhere, *is* just: a hierarchy + a downward-inherited "keeps-apart" relation + overlap as the leftover. The "$\Rightarrow$" is a 150-line kernel proof (`propext` only); the "$\Leftarrow$" is GPT-5.6 Pro's canonical set representation, also kernel-checked for arbitrary domains — the structural fact both major proof routes silently relied on, now a certified biconditional.

A late surprise: the problem's exact address is Π^0_1



- The whole semantics is finitely first-order axiomatizable; by Gödel completeness, *un*satisfiability is enumerable.
- So SAT is co-r.e. (Π^0_1) — the domino problem's *ceiling* (membership; we don't claim hardness).
- Nobody in the project — or the reviews — had noticed for four months. Machine-cross-checked; the reduction is in Lean.

Consequence: half of the keystone was never needed

A Π^0_1 problem is decidable *iff* SAT is also enumerable. So run two search parties and wait for the first hit:

Party A: enumerate finite certificates
checker verified in the kernel (`finAccept_sound`)

Party B: enumerate first-order refutations
complete by Gödel's theorem (classical, citable)

?

first hit decides

- Needed now: only qualitative F6 — *some* finite certificate per satisfiable concept, no computable bound.
- The bound the reviewers kept demanding (and that kept breaking) comes back for free, afterwards — a theorem of the schema.

“Isn’t this just first-order logic?” — yes, and it helps

$\mathcal{ALCI}_{\mathrm{RCC5}}$-SAT *is* FO satisfiability of a finite theory. Does it land in a decidable fragment of FO?

Fragment	Fate	Why
two-variable FO^2	$\times$	composition is ternary (RCC3 *did* fall this way in 2003!)
prefix classes	$\times$	unbounded $\forall\exists$ alternation, FMP
guarded / clique-guarded	$\times$	totality is the one unguardable axiom
unary/guarded negation	$\times$	finite model property; totality again
guarded + transitivity	$\times$	undecidable outside guards

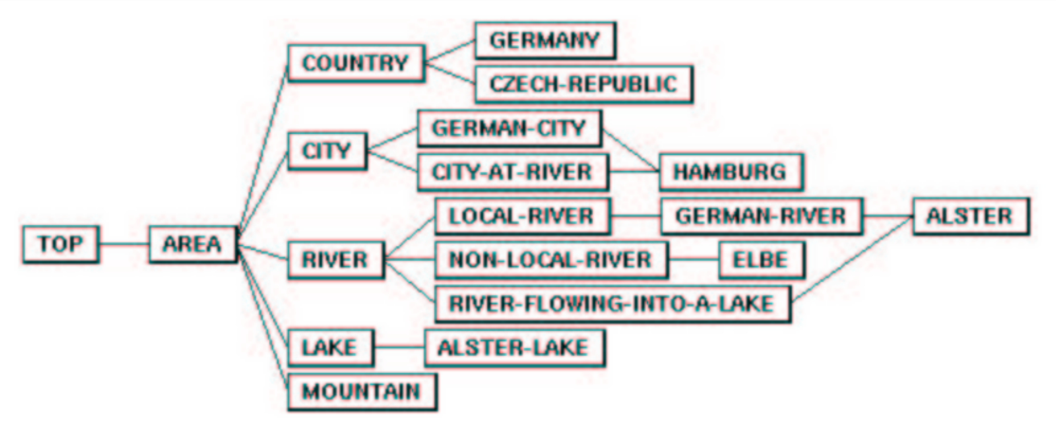
Every failure traces to the same fact: every pair is related, so models are maximally non-local — exactly the obstruction F6 names. The FO view rediscovers the project’s wall from classical model theory.

A working prototype reasoner (validated, not proved)

- The cover-tree tableau ($\sim$350 lines of Python): exact-occurrence blocking on the split-forest foundation.
- Cross-validated against an independent oracle on 911 concepts — zero mismatches; 775/775 structured decomposition tests; never once contradicted in the whole campaign.
- Decides correctly even the known trap (the strong-EQ PO-loop) that fools type-based procedures.

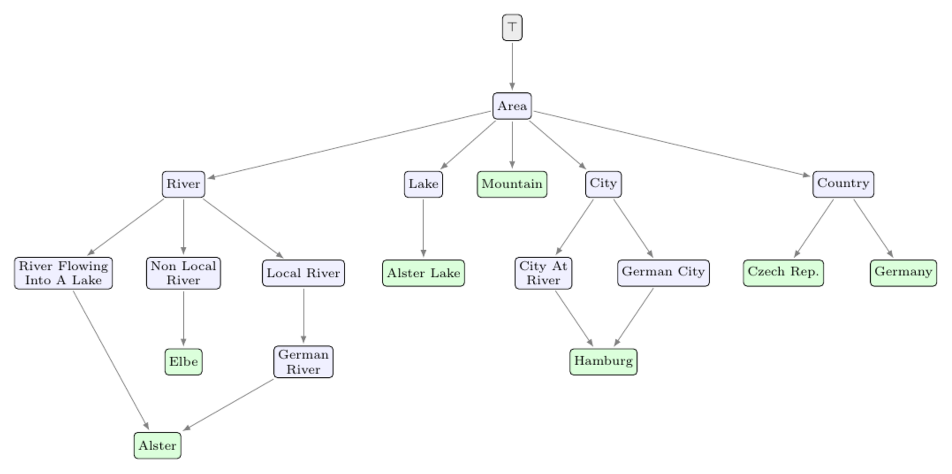
A running algorithm and a kernel-checked proof, pointing the same way — the strongest evidence short of a certificate.

2003: a prototype reasoner



The GIS concept taxonomy in report7 (Fig. 6), 2003 — itself *computed by a prototype reasoner* of the era, not drawn by hand.

2026, the cover-tree reasoner: all 21/21 — edge for edge



The 2026 cover-tree tableau reproduces it exactly — two different reasoners, 23 years apart, identical result (green = individuals; note *Hamburg* and *Alster* each have two parents — the

What the models did — and what they could not

Demonstrated

- months-long rigorous formal development, $\sim$30 rounds
- genuinely useful *adversarial* review (with machine-checked witnesses)
- a 5,900-line kernel-checked Lean development
- working, validated software

Not demonstrated

- the decisive novel idea — F6 is uncracked
- reliable *self*-review (both models declared victory prematurely; both were caught)
- working unsupervised: the human directed every round

Neither model is self-sufficient. What carried the project was the loop, not any single model.

The method that worked (steal this)

1. **Cold review beats hot review.** A *fresh* session with no memory of the construction found what warm reviewers missed — every time it mattered. Make it a standard step.
2. **Don't trust — verify.** Every load-bearing claim was cross-checked mechanically: 88 probes that re-derive the algebra from set semantics, and ultimately the Lean kernel. The machine checks are the ground truth, not the model's confidence.
3. **Label honestly.** *Strongly supported* $\neq$ *certified*. Overclaiming happened twice; cold review caught both; the claims were withdrawn — on the record.

The full timestamped conversation log is public — every idea, defect, and repair traceable to its origin, human or model.

If nothing else: a message to fellow researchers

- Whatever the fate of the mathematics, this campaign is an existence proof of what frontier models could already do in serious formal research — *in July 2026*.
- Anyone doing serious logic or mathematics should know these capabilities exist. Ignoring them does not advance science; declining to use them is simply unwise.
- Saying exactly this was a stated purpose of the work. Its first outing (DL 2026) was rejected without engaging either the mathematics or the message — so it is now on arXiv, in the open, with the full audit trail.

The instruments have changed. The standards — verification, adversarial review, honest labels — matter more than ever.

The results, by verification level

Level	Results	What it means
Lean-certified	soundness pipeline; abstraction faithfulness; RCC5 normal form (both directions); ∀PO-free lift core; F6 $\Rightarrow$ decidable; Π^0_1 dovetail	kernel-checked, with a stated axiom footprint
Theorem-level	∀PO-free decidability + bound; identity-selector characterization; W2′ folds into F6	proof in a work package, often with probes
Empirical	cover-tree tableau; 911-concept cross-validation; model verifier	test evidence, not a decision theorem
Open	static F6; completeness of the certificate scheme; automatic-cover property; full decidability	no proof claimed

Four levels kept rigorously distinct — conflating them is the easiest way to misread the project.

Where it stands

One well-posed question

can a concept force unbounded *identity-selector* structure?

NO $\Rightarrow$ decidable (machinery ready, kernel-checked) YES $\Rightarrow$ the grid/undecidability route opens

- 20+ years of difficulty, compressed into one lemma — with certified machinery waiting on both sides of the answer.
- Everything is frozen, tagged, reproducible: paper (42pp), 4 Lean files, 88 probes, 2 reasoners, full conversation log.

Open since 2003. Smaller than ever. Still standing.

`github.com/lambdamikel/alcircc5` — thank you!

Backup: the full RCC5 composition table

comp	DR(a, b)	PO(a, b)	EQ(a, b)	PPI(a, b)	PP(a, b)
DR(b, c)	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
PO(b, c)	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
EQ(b, c)	DR	PO	EQ	PPI	PP
PP(b, c)	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
PPI(b, c)	DR	DR, PO, PPI	PPI	PPI	$*$

Row $= S(b, c)$, column $= R(a, b)$, entry $=$ possibilities for (a, c); $* =$ all five. Re-derived from finite set semantics by every probe; matches the certified Lean table cell for cell.

Backup: what a finite certificate looks like

- **Catalogue** (the finite blueprint):
  - a *core network* — the non-repeating part of the model;
  - finitely many *attachment templates* — recipes for gluing a fresh witness cluster onto an interface;
  - a *steering function* fixing the horizontal relations from old occurrences to fresh ones (the only real freedom).
- **Plan**: which template to apply where; *unfolding* it (possibly forever) produces the split-forest model.
- **Soundness**, kernel-checked: one finite check on the catalogue guarantees *every* unfolding is a legal RCC5 model.
- **Live width** = live occurrences that must coexist at one interface during unfolding — the quantity F6 must bound.

Backup: the sharpened F6 — identity-selector minors

- The naive ways to force width (shell half-graphs, prefix pumps, PO-gaps) all collapse to finite catalogues — machine-checked.
- The sharp obstruction: n pairs (L_i, U_i), each kept apart by a private selector A_i and by nothing else — the selector incidence matrix is the $n \times n$ *identity*.
- A finite catalogue can hold only boundedly many private selectors, so: **F6 fails iff some concept forces such minors for every n.**
- Bare RCC5 networks with unbounded such structure *exist* (machine-checked) — so any F6 proof must use $\mathcal{ALCI}_{\mathrm{RCC}}$*-forceability*, not RCC5 consistency alone. The uniformization side-condition (W2$'$) provably folds into F6.